MUHAMMED BERK ÖNDER

AI/ML Engineer

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AI/ML Engineer with experience building, deploying, and scaling production ML and LLM systems in Telecommunications and Software/Mobile Applications industries. Specialized in Anomaly Detection, large-scale data processing, and LLM-powered systems and data pipelines, with strong expertise in Python, Docker, Git, Airflow, GCP, and modern LLM frameworks (LangChain, Ollama).

Experience

MLOps Engineer *Teknasyon* 08/2024 - Present

- Designed and deployed a modular RAG-based LLM system using LangChain and Open WebUI, enabling high-precision retrieval from structured internal data via custom Python pipelines and tool integrations.
- Scaled a production spam detection pipeline of millions of weekly records by segmenting data processing, eliminating memory bottlenecks and improving stability.
- Built automated data pipelines for anomaly detection using Dataform and Airflow on GCP, enabling scalable ingestion of high-cardinality event data and reducing manual query overhead.
- Integrated MLflow into Airflow pipelines to standardize experiment tracking, improving model observability and reproducibility.
- Optimized fine-tuning of ~30B parameter Qwen models on RunPod, reducing training time and infrastructure costs through better GPU utilization.
- Implemented CI/CD pipelines with GitHub, Cloud Build, and Docker to produce versioned model artifacts and enable reproducible deployments.
- Processed high-cardinality event data across thousands of dimensions, improving pipeline scalability and reliability.
- Containerized ML and LLM services with Docker for consistent, portable deployments across environments.

Machine Learning Engineer *P.I. Works* 07/2022 - 07/2024

- Built a telco LLM assistant using LangChain, enabling knowledge base retrieval, SQL-based data access, analysis, and visualization within a unified interface.
- Developed a heatmap forecasting system using regression models to predict network traffic from football match data, reducing aggregate error from 77% to 0.75%.
- Implemented multivariate anomaly detection on streaming time-series data using clustering techniques, improving detection across diverse scenarios.
- Designed a synthetic data generation framework using Monte Carlo methods, increasing data granularity by 60×.

Education

İstanbul University

MSc, High Energy and Plasma Physics GPA: 3.63 Graduation Date: 2024

BS, Physics GPA: 3.31 Graduation Date: 2021

Skills

Programming: Python, SQL, Bash, VIM, Tmux

ML/LLM Frameworks: Scikit-Learn, MLflow, TensorFlow, PyTorch, PySpark, Huggingface, LangChain, LangGraph, Ollama, LMStudio, vLLM, Unsloth

Python Packages: Pandas/Polars, NumPy, Matplotlib, Plotly, Typer, NetworkX

MLOps & Infra: Git, Docker, CI/CD, FastAPI, Airflow, GCP, Linux, uv, Nix, Prefect, RunPod, Podman

Languages: Turkish (native), English (proficient)